

Python Developer | AI/ML Enthusiast | Computer Engineering Student

h-edgethedevelop My portfolio barberheritage.netlify.app

Computer Engineering student combining software development, machine learning, and renewable energy systems. Currently building projects in Python and exploring AI applications in real-world problem solving.

CapTech Cohort II Team Lead/Frontend Developer, Sara Foundation	04/2026 – 05/2026
Frontend Developer and Team lead. Built a project - Studypath AI for enhanced learning	

BSc Computer Engineering, *University of Lagos* 10/2024 – Present
Lagos, Nigeria

Python Programming

- Data types, loops, and functions
- Object-Oriented Programming (OOP)
- File handling and exception management
- PyQt5 & Flet frameworks
- Problem-solving and algorithmic thinking

C/C++

Competent in writing structured, efficient C/C++ programs; experienced with pointers, dynamic memory, arrays, structs, file I/O, and modular code design. Able to solve medium-level problems and implement core algorithms.

Git & Version Control

Repository management and branching workflows, Committing, pushing, and maintaining open-source projects on GitHub• .gitignore configuration and PowerShell-based Git workflows on Windows

Leadership/Initiative

Proactive and driven, able to lead small projects or initiatives and motivate peers toward goals.

AI Prompt Engineering

Ability to construct concise prompts in order to get desired results from AI models

JavaScript(Frontend)

- Core JavaScript syntax (variables, loops, arrays, functions)
- Object-Oriented Programming (classes, constructors, methods)
- DOM basics and event handling
- HTML & CSS integration for interactive web pages

Artificial Intelligence and Machine Learning

Gained practical experience in Python-based machine learning; implemented small projects involving data preprocessing, model training, and evaluation. Familiar with basic algorithms such as linear regression, classification, and decision trees.

Critical Thinking and Problem solving

Ability to profer solutions to complex tasks and problems, able to break down complex issues and implement effective solutions.

Adaptability / Learning

Quick learner, adaptable to new technologies and challenges in dynamic environments

Photo-voltaic Engineering

Basic ability to perform Energy Audits, Panel sizing and installation of solar panels

- English
- Yoruba
- French

Certificates

- Certificate of Completion: Python Essentials 1- Cisco Networking Academy
- Certificate of Completion: AI Literacy - IBM SkillsBuild [↗](#)
- Certificate of Completion: C++ Training Crash Course - Udemy
- Certificate of participation - Claude Builders club Hackathon
- Certification of Completion: AI for Beginners course - HP Life
- Certificate of Completion: Prompt Engineering with ChatGPT
- Certification of Participation in Solar Design & Installation -Intelligent Systems Lab, University of Lagos.
- Certificate of Participation - CAP Cohort II [↗](#)

Awards

- Best in Common Entrance, Mountain Top Primary School** 08/2018
Award given to the best participant of the common entrance examinations
- Award of 100% scholarship,** 12/2023
Jubril Martins Memorial Grammar School old students association
100% Scholarship given to the best participating students and a laptop
- Award for Love, Dedication and Support, Xtralarge Farms** 09/2024

Projects

- ASL-to-Text/Speech Conversion Model** 02/2026 – Present
Building a computer vision model as a second-year Computer Engineering student that recognizes American Sign Language gestures and converts them to text and speech. Developed using Python, PyTorch, and Scikit-learn, with a Streamlit interface for real-time interaction. Aimed at improving accessibility for the deaf and hard-of-hearing community.
- Voltmart E-Commerce Website** 01/2026 – Present
Designed and developed a gadget e-commerce platform using Python and the Flet framework, featuring a clean UI for seamless product browsing and sales. Built entirely in Python, demonstrating ability to deliver full frontend experiences without traditional web technologies.

Publications

- INVESTIGATION OF WIRELESS COMMUNICATION SYSTEMS USING WAVELET ANALYSIS IN DEFENSE TECHNOLOGIES** [↗](#), Barber Heritage Oluwanifemi 29/01/2026
This comprehensive literature review highlights the advances in wavelet transforms in signal edge detection, spectrum analysis, and so on. A conceptual solution is proposed and a Python based wavelet sensing algorithm proves that wavelet analysis can effectively detect jamming signals hence enhancing the efficiency of defense communication systems.

Organisations

- SEES Innovation Hub, Asst. Team lead - Welfare and Logistics team** 06/2026 – Present
Lagos, Nigeria
- Claude Builders Club, Member** 03/2026 – Present
Lagos, Nigeria
- Xtralarge Farms, Volunteer Graphics Designer** 01/2021 – Present
Lagos, Nigeria